

The Job Sequencing and Tool Switching Problem

Consolidated Open Problems and Their Current Status

This is an index of the open problems raised across Parts I–VII, with precise statements, current status, and pointers. “Verified” means checked by the scripts in `plans-genai/_verification/` (see `VERIFIED_FACTS.md`). Conjectures are labelled as such; nothing here is claimed as proved unless it is proved in the cited Part.

Gap and Approximation Ratio (Part V)

OP 1 (Tight ratio for $b = 3$; conjecture). Is $H/Z_{\text{SSP}}^* \leq 4/3$ for every $b = 3$ instance, with equality only at the 6-job ring? *Status:* open. Strong evidence — no counterexample in the 10691 exhaustively enumerated $b = 3$ edge instances, nor in a further 6065 instances with size-3 jobs, nor in the copy-paste or ring families (Part V; $\approx 17\text{k}$ instances total). The earlier route via $\sum_k R_k \leq H/4$ is *false* (counterexample $\{a, b, c\}, \{a, b, d\}, \{c, d, e\}$) and has been retracted.

OP 2 (Extremal ratio for general b). What is $\sup_I H/Z_{\text{SSP}}^*$ as a function of b ? *Status:* partially resolved. Exact witnesses give ratios $4/3$ ($b = 3$), $4/3$ ($b = 4$), and $3/2$ ($b = 5$). These are *proved* lower bounds on $\sup_I H/Z_{\text{SSP}}^*$ (the instances and their Z^*, H are computed exactly), so it is a **theorem** that $\sup_I H/Z_{\text{SSP}}^* > 4/3$ for $b \geq 5$ — i.e. the $4/3$ bound is specific to $b = 3$, *not* universal. What is *not* proved (three data points only): monotonicity in b , the growth law (does it $\rightarrow 2$? is it unbounded?), and the $b = 3$ upper bound itself (OP 1).

OP 3 (Structural characterisation of zero gap). Give a necessary-and-sufficient structural condition for $H = Z_{\text{SSP}}^*$. *Status:* open, partly clarified. By OP 4 the gap vanishes for *every* instance once grouping choice is unrestricted, so the question is really: when does some *minimum-cardinality* (K^* -)grouping already attain Z_{SSP}^* ? Two sufficient conditions are proved (the heuristic meets either lower bound $K^* - 1$ or $|U| - b$; Part V), and empirically every positive-gap $b = 3$ edge instance has $K^* = 3$ with the ring/prism boundary structure. A clean necessary-and-sufficient characterisation (e.g. via a forbidden “ring obstruction”) is missing.

Grouping Selection (Parts V–VI)

OP 4 (Joint grouping–sequencing). *Structural part now resolved* (Part V, grouping-exactness theorem): $Z_{\text{SSP}}^* = \min_{\mathcal{G}} \text{GTSP}(\mathcal{G})$ over *all* feasible groupings, so the JGP+GSP gap is *exactly* the price of restricting to minimum-cardinality groupings (proved; verified on 1306 instances). In particular a sub-optimal grouping ($K > K^*$) can strictly beat the heuristic (the 6-ring $K = 4$ grouping attains $Z_{\text{SSP}}^* = 3 < 4 = H$). *What remains open* is the

algorithmic problem: efficiently optimise over the (exponentially many) feasible groupings — a polynomial algorithm, or NP-hardness, for this joint grouping-and-sequencing objective.

OP 5 (Beyond one-directional lookahead). The MWHP ordering result (Part VI) is exact only for the one-directional lookahead policy and coincides with the optimum at zero slack; with positive slack, bidirectional padding can be strictly cheaper (the sliding-window family has true cost $|U| - b = (K-1)(r-\text{ov}) - (b-r)$, below the MWHP value $(K-1)(r-\text{ov})$). A polynomial exact GSP for positive-slack groupings, or a proof of NP-hardness, is open.

Exact Formulations (Parts II, IV)

OP 6 (Exchange argument / cluster-MTZ exactness; central to Part IV). RESOLVED (negative). The cluster-aggregated MTZ formulation is *not* exact. Its constraint is representable for a configuration path *iff* every ordered cluster pair is traversed at most once *and* the induced precedence digraph (edge $j' \rightarrow j''$ whenever an arc runs from a j' -config to a j'' -config) is *acyclic*. The acyclicity fails under **job domination**: if $T_i \subseteq T_j$, every config covering j also covers i , so an arc between two such configs forces both $j \rightarrow i$ and $i \rightarrow j$, a 2-cycle. Explicit counterexample ($b = 3$): jobs $\{1, 4\}, \{1, 3, 4\}, \{1\}, \{1, 2, 4\}$ have $Z_{\text{SSP}}^* = 1$ but cluster-MTZ optimum 2 (brute-force verified; 19 counterexamples in $\approx 7.9\text{k}$ instances). **For column generation use per-configuration MTZ or lazy SEC/PCTSP cuts (both exact), not cluster-MTZ.** The job-indexed MTZ of the BBC master (Part VIII) is unaffected.

OP 7 (LP strength and net solver benefit of MTZ). The cluster-MTZ relaxation is no tighter than SEC and is weakened further by aggregation (Part IV). Quantify the LP gap on SSP instances (with and without Desrochers–Laporte lifting), and measure whether removing row generation yields a net branch-and-price speed-up versus the SEC and SSPMF formulations on the Catanzaro Tabela1C set. *Status*: open, computational.

OP 8 (Fractional Benders cuts). The branch-and-Benders solver (Part VIII, 08_branch_and_benders.tex) currently adds optimality cuts only at integer master solutions (KTNS subproblem). Deriving valid cuts at *fractional* master solutions from a relaxation of the KTNS recursion (min-cost-flow / interval-matrix lower bound, or the totally-unimodular tooling LP) — enabling user cuts that tighten the bound before branching — is open. Cut *lifting* (certifying a cut for all sequences sharing a 0-block structure) is the related practical lever.

Cost-Function Variants (Part VII)

OP 9 (Tight collapse condition). The conditions (C1)+(C2)+(C3) are sufficient for collapse of an SSP variant to the JGP; the necessity analysis (Part VII) shows (C3) necessary and (C1),(C2) dispensable. A single necessary-and-sufficient characterisation of collapse, beyond the reduction to the threshold $\rho_c(I)$, is open.

OP 10 (Setup-time regime in practice). For the magazine/tool setup-time ratio ρ , the TD-SSP approximation ratio $R(\rho)$ decreases from $\leq 4/3$ at $\rho = 0$ to 1 as $\rho \rightarrow \infty$, with ring

threshold $\rho_c = 1$ (Part VII). Whether the manufacturing regime $\rho \in [3, 60]$ (Privault–Finke) makes JGP+GSP exactly optimal on broader instance classes (not just ring-like) is open.

Cross-references: *OP 1–4* (Part V, 05); *OP 5* (Part VI, 06); *OP 6–7* (Part IV, 04); *OP 8* (Part VIII, 08); *OP 9–10* (Part VII, 07).